HAMZA HAMID

AI Product Engineer · Full-Stack & Applied AI

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Profile

Full-stack engineer who designs and ships production AI and blockchain systems end to end. Five years across TypeScript/Node, Python, React, Solidity and Move wiring large language models and generative-media APIs into reliable, queue-based backends, and architecting institutional-grade tokenization and compliance infrastructure. I build the whole stack: data ingestion, model and contract orchestration, job pipelines, cloud automation, and the front-end that makes it usable. I bring rare commercial context for regulated and Gulf markets, having led real regulatory engagement across DIFC, ADGM, VARA and MiCA.

Professional Experience

Senior Blockchain Engineer — Mobius Network Artificial Intelligence 02/2025 – Present
Dubai, UAE · Real-world asset (RWA) tokenization across global jurisdictions

- Lead end-to-end architecture, regulatory alignment and technical execution for RWA tokenization, contributing directly to product strategy, compliance and institutional adoption.
- Designed institutional-grade RWA architecture using **ERC-3643, ERC-1400, ERC-3475** and custom compliance modules.
- Built issuer, investor and marketplace portals: **KYC/AML engines, wallets, vaults, SPV structures, cap-tables** and automated compliance workflows.
- Managed full regulatory engagement across **DIFC, ADGM, VARA, DLD, MAS (Singapore) and MiCA (EU)** for issuance, secondary trading and investor onboarding.
- Integrated **Chainlink oracles, Tokeny compliance frameworks, Ondo liquidity models**, custodians, identity systems and permissioned chains.
- Structured SPVs with global legal partners (Aurum, Baker McKenzie) ensuring regulatory alignment, audits and investor protection.
- Led GTM strategy for tokenized offerings, fractionalization models, liquidity pathways, stablecoin rails and settlement layers and built multi-role issuer↔investor↔admin dashboards with real-time reporting.
- Collaborated with institutional investors, family offices and DeFi funds on early-stage tokenized listings; managed licensing, tax, accounting and government filings.
- Represented Mobius at the **Origins x Token2049 Singapore Hackathon**, coordinating with global founders and protocols.

Full-Stack Web Developer — Emirates National Schools 02/2023 – Present
Abu Dhabi, UAE

- Develop and maintain the **Emirates National Schools (ENS) website** and internal web platforms in React and TypeScript, ensuring performance, security and reliable day-to-day operation.
- Built **student-data automation** pipelines that streamline admissions, records and reporting, reducing manual workload and improving data accuracy across departments.
- Engineered cloud infrastructure on **Microsoft Azure** and integrated **Dynamics 365** for CRM, records and workflow automation across the institution.
- Developed **AI automation bots** for support, notifications and routine administrative tasks, integrating messaging/SMS systems for secure communication.
- Delivered payment-gateway integration for fee management and a multi-department intranet using Docker and Kubernetes.

Blockchain Developer — Kryptohive / Pontem Network 08/2022 – 10/2023
Islamabad, Pakistan

- Built the **Pontem Wallet** front end in React wallet infrastructure that grew to **1M+ active users** on Aptos, one of the most popular wallets on the network.

- Wrote and audited **LiquidSwap** DEX smart contracts in Move, reducing swap transaction costs ~10%.
- Improved wallet UI/UX and supported a token launch that drove 200K+ active users; contributed to an Ethereum↔BSC cross-chain bridge.

Full-Stack Developer — XP Network 08/2021 – 02/2023

Remote

- Collaborated with product and engineering teams to improve platform efficiency ~25%.
- Engineered 5 reusable cross-chain libraries, reducing the codebase ~10%.
- Built NFT-transfer tooling and the foundational backend for the XP Network NFT bridge and FinTech integrations.

Frontend Developer — Precise Tech 02/2020 – 08/2021

Remote

- Developed a custom e-commerce platform for a fashion brand, contributing to a ~30% increase in online sales.
- Built a real-time chat application for a financial-services company for instant customer support.
- Designed a scalable cloud architecture for a healthcare startup, improving system reliability and uptime ~50%; audited DeFi and NFT smart contracts.

Key Projects & Products

Engineer & Builder — Qumak (qumak.io) 2026 – Present

AI marketing-intelligence & ad-generation platform for Gulf SMBs and agencies

- Built, independently, an **end-to-end URL-to-Ads pipeline**: brand URL → automated multi-page scraping → brand-kit/palette/social extraction → competitor ad intelligence → AI-generated image and video ads.
- Engineered a **multi-stage video-ad system** that decomposes an ad into a shot recipe, generates each shot via generative-video models (fal.ai – WAN, Kling, Seedance), syncs results through a worker, and assembles a branded ad with FFmpeg; added multi-reference conditioning for character/product consistency across clips.
- Architected on **Node/Express, MongoDB, BullMQ/Redis, Cloudflare R2**; integrated Stripe billing and an LLM intelligence engine with anti-fabrication prompt controls.

Engineer & Builder — Tammat (tmmt.ae) 2025 – Present

AI-powered UAE visa & document-processing platform

- Built an **AI assistant for UAE visa and residency processing** combining LLM chat with a structured government-service catalogue and human-officer escalation.
- Reached **AED 15K MRR and processed 214 applications**; owned full stack — React front end, Node API, Stripe payments and the AI conversation layer.

Technical Skills

AI / ML: LLM orchestration & prompt engineering, generative video/image APIs (fal.ai — WAN, Kling, Seedance), multi-reference conditioning, AI automation bots, Azure TTS, anti-hallucination prompt design

Backend & Cloud: Node.js, Express, Python, MongoDB, Redis/BullMQ, Microsoft Azure, Dynamics 365, Stripe, Cloudflare R2, Docker, Kubernetes, FFmpeg, GraphQL

Frontend: TypeScript, React, Next.js, Vite, React Native, Vue.js, Tailwind CSS, Three.js, Figma

Blockchain / Web3: Solidity, Move, Hardhat, Ethers.js, ERC-3643/1400/3475, Chainlink, Tokeny, cross-chain bridges, smart-contract auditing (Aptos, Ethereum, Polygon, BSC)

Regulatory / Domain: RWA tokenization, DIFC/ADGM/VARA/DLD/MAS/MiCA engagement, KYC/AML, SPV structuring, UAE licensing & PRO processing

Education & Certifications

- **B.Sc. Computer Science — Quaid-e-Azam University, Islamabad**
- Origins x Token2049 Singapore Hackathon – participant (2025); presented Tammat AI
- Certified Blockchain Expert (Blockchain Council) · Solidity Smart Contract Development (CryptoZombies)

Languages

English (IELTS 6.0) · Hindi/Urdu (native) · German (A2)